

C++ in Quantitative Finance part #2

Individual Home Project for: Antoni Smolski (473975)

1st deadline: 2026-06-15, 23:59:59

2nd deadline: 2026-08-31, 23:59:59

You are considering a **long put butterfly** option strategy for **European options** with **barriers**. Such strategy is based on a portfolio consisting of one long OTM put, two short ATM puts, and one long ITM put. All options have the same time to maturity. Assume that call options (if you have any) are of down-and-out flavour and that put options (if you have any) are of up-and-out flavour.

Specifically, you are interested in the relationship between the theoretical price of this portfolio and three factors (simultaneously): (1) price of the underlying (after the portfolio is constructed), (2) volatility, and (3) level of the barrier.

Your task is to:

1. Modify the pricer developed in `project11/project12` and perform Monte Carlo simulations which will find approximations of prices of appropriate options and illustrate this relationship:
 - **either in Excel:** using the `xlw` wrapper software to import your C++ function to MS Excel. Basically, calling the function for appropriate options with given spot prices and maturities should be enough. Create appropriate plots which will visualize the relationship described above in a clear and readable way.
 - **or in R:** as above, but using the `Rcpp` package. Build your own **R source package**, install and load the package in the script in the separate R project, call appropriate function/functions to find out prices of options, and visualize the relationship with appropriate plots.

Include in your plots the appropriate range of analyzed factors.

Hint: do not overcomplicate things! Remember that value of the portfolio consisting of two options is given by the sum of values of single options (with appropriate signs, according to the position of the option in the portfolio).

2. Write a report on it.

Your code should be ready to price options for any values of their characteristics. Nevertheless consider the following values for options parameters:

- price of the underlying at the moment of option pricing: $S_0 = 110$
- annualized volatility rate $\sigma = 23\%$
- annualized risk-free rate $r = 5\%$
- annualized dividend rate $d = 0\%$
- time to maturity $t = 6$ months

RULES

1. Each project is to be prepared **individually**.

2. Use comments in your code as much as possible.
3. Your submission should consist of the following elements:
 - (a) one zip file which includes
 - either: the whole `xlw` solution (see below) together with the resulting `xll` file and the `xls/xlsx/xlsm` file with the application of the function to the problem,
 - or: a complete **R source package** which is ready to be installed (see below) and a separate R project with a script or Rmd file with the application of the developed function from the package to the problem (simulations visualisations, etc.),
 - (b) a `pdf` file with a short report (see below).
4. Rules to prepare the `zip` file with the `xlw` solution and the `xls/xlsx/xlsm` workbook:
 - (a) Double-check if your code compiles without errors.
 - (b) Add your `xls/xlsx/xlsm` Excel workbook with the application at the top of your `xlw` project folder.
 - (c) ZIP it all.
5. Rules to prepare the **R source package** and application of the R function:
 - (a) Produce a **source package** using the `Rcpp` package as discussed during `cpp07` classes.
 - (b) Add next to the package (`*.tar.gz` file) another R project with the R script file or the `Rmd` file with the application of the developed function to the problem.
 - (c) ZIP it all (the package and the R project with the script/`Rmd` file with the application).
6. Your report (a `pdf` file) should include short information about:
 - (a) the objective of the project,
 - (b) assumptions (for example: dynamics of the prices of underlying, number of iterations in Monte Carlo, etc.),
 - (c) the portfolio to be priced and its characteristics,
 - (d) code elements used (short explanations),
 - (e) results of your study: prices of the portfolio, a plots representing the relationship and your comments to that.

Please note, that the report is a very important part of your submission. Report of bad quality may significantly contribute to your overall score.
7. Your submission files should have following names: `XXXXX.zip` and `XXXXX.pdf`, where `XXXXX` is your surname. Examples: `sakowski.zip`, `sakowski.pdf`.
8. Share your solution (one `zip` file and one `pdf` file) with `p.sakowski@uw.edu.pl` via Google Drive, OneDrive, Dropbox or similar services with a clickable link. You should get a confirmation of this from me within a couple of hours.
9. Include in your report a following declaration:

In accordance with the Honor Code, I certify that my answers here are my own work based on class materials. I did not make my solutions available to anyone else, and that I did not engage in any activities that would dishonestly improve my results or improve or harm the results of others.
10. Include in your report a brief statement indicating whether you used any AI tools. If you did, briefly describe the extent of their use.

11. Cheating is a very serious offence. Please keep in mind that any evidence of copying of your home project solutions (the report as well as the code) will be treated as cheating and will be reported immediately to the Dean. Please take it seriously.

Good luck!

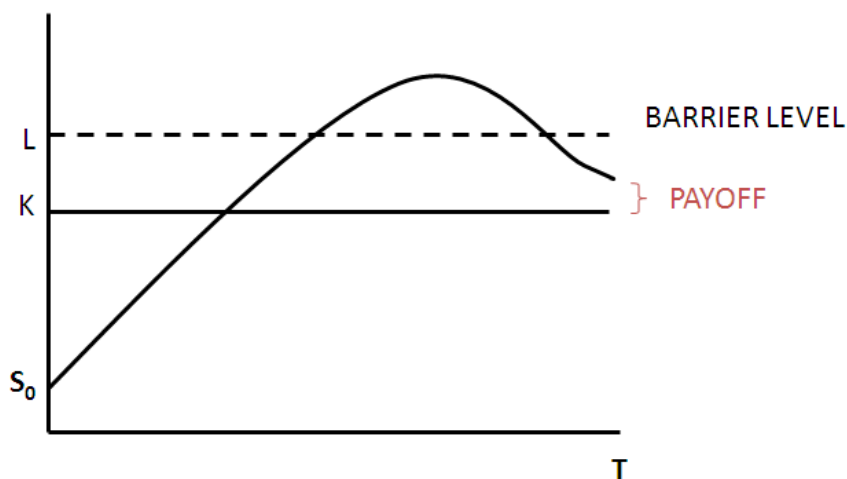
Paweł

A short refresher: payoffs of EUROPEAN barrier options with continuous barriers

1. CALL UP-and-IN

- the price has to increase to reach the barrier
- activated when asset price reaches the barrier

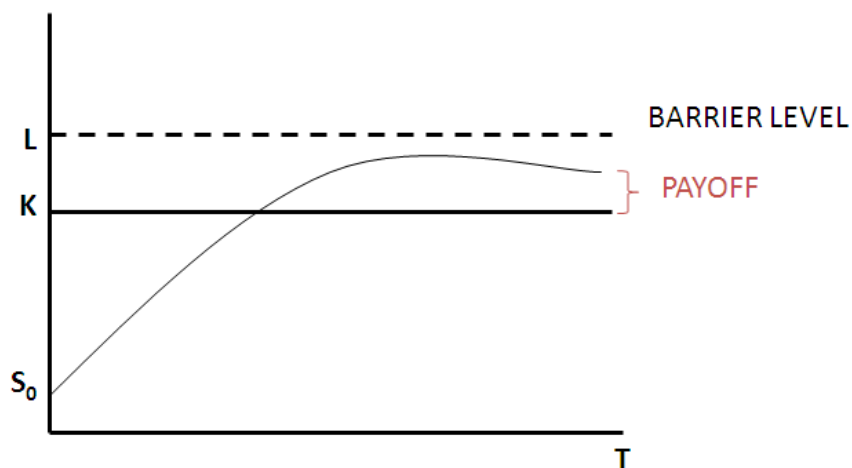
$$\Phi_t = \max(S_T - X, 0) \quad \text{if} \quad \max_{0 \leq t \leq T} (S_t) \geq L \quad (1)$$



2. CALL UP-and-OUT

- the price has to increase to reach the barrier
- canceled when the asset price reaches the barrier

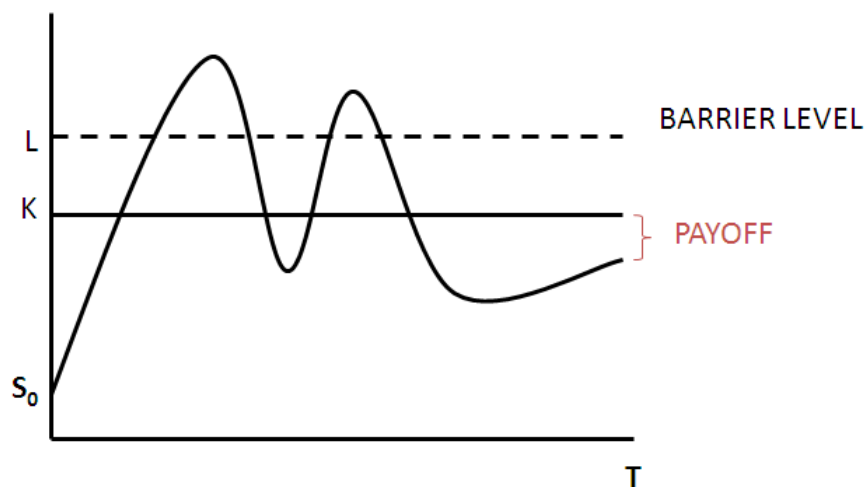
$$\Phi_t = \max(S_T - X, 0) \quad \text{if} \quad \max_{0 \leq t \leq T} (S_t) \leq L \quad (2)$$



3. PUT UP-and-IN

- the price has to increase to reach the barrier
- activated when asset price reaches the barrier

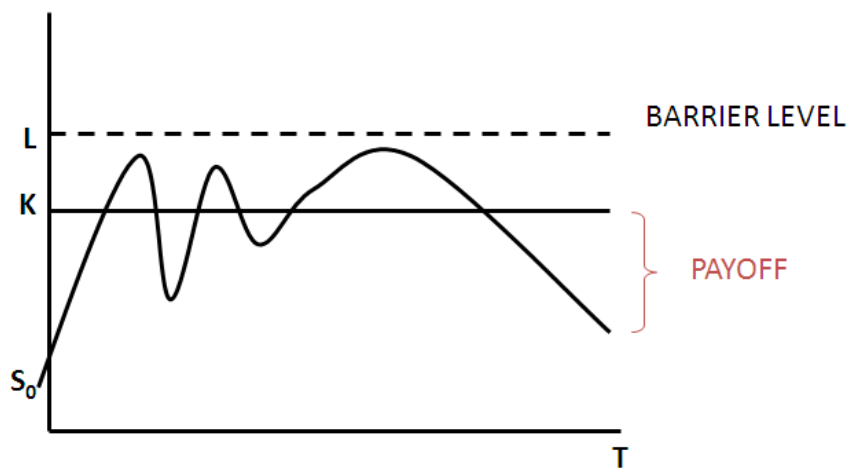
$$\Phi_t = \max(X - S_T, 0) \quad \text{if} \quad \max_{0 \leq t \leq T} (S_t) \geq L \quad (3)$$



4. PUT UP-and-OUT

- the price has to increase to reach the barrier
- canceled when the asset price reaches the barrier

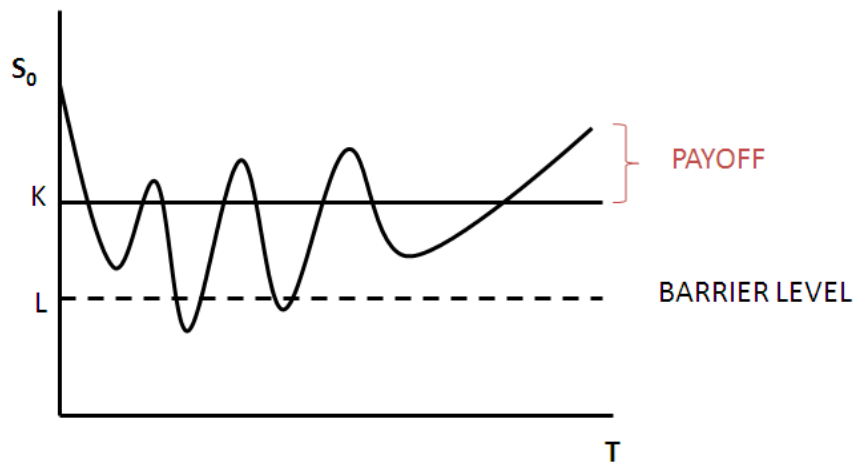
$$\Phi_t = \max(X - S_T, 0) \quad \text{if} \quad \max_{0 \leq t \leq T} (S_t) \leq L \quad (4)$$



5. CALL DOWN-and-IN

- the price has to decrease to reach the barrier
- activated when asset price reaches the barrier

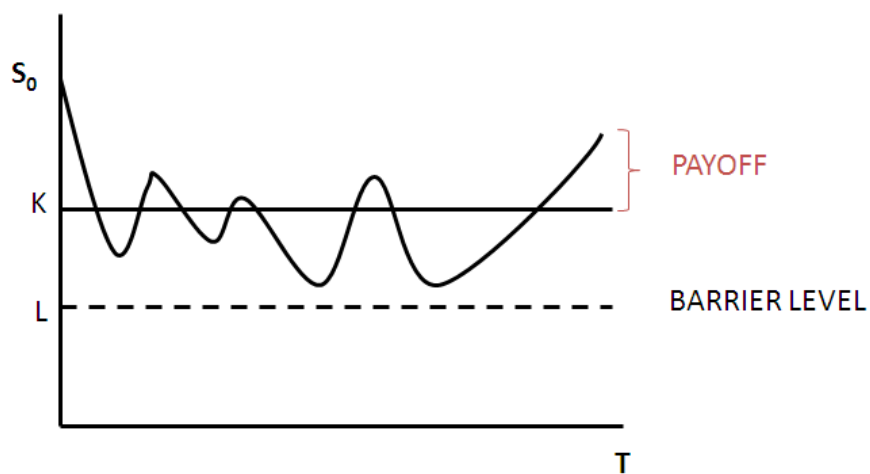
$$\Phi_t = \max(S_T - X, 0) \quad \text{if} \quad \min_{0 \leq t \leq T} (S_t) \leq L \quad (5)$$



6. CALL DOWN-and-OUT

- the price has to decrease to reach the barrier
- canceled when asset price reaches the barrier

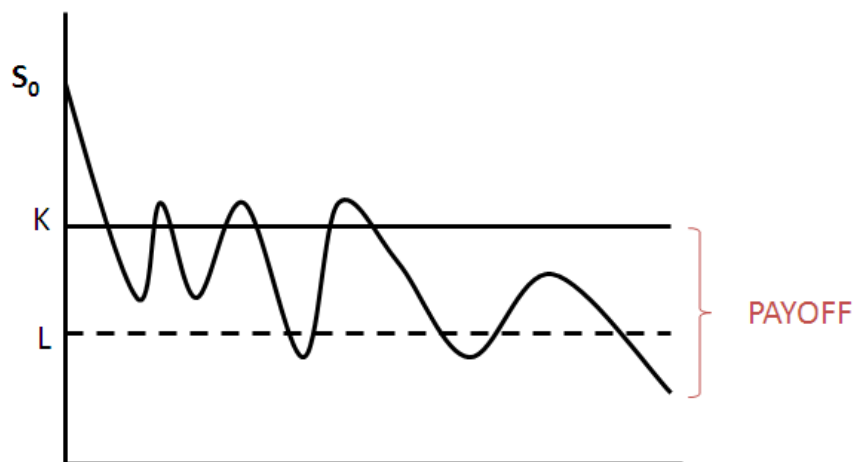
$$\Phi_t = \max(S_T - X, 0) \quad \text{if} \quad \min_{0 \leq t \leq T} (S_t) \geq L \quad (6)$$



7. PUT DOWN-and-IN

- the price has to decrease to reach the barrier
- activated when asset price reaches the barrier

$$\Phi_t = \max(X - S_T, 0) \quad \text{if} \quad \min_{0 \leq t \leq T} (S_t) \leq L \quad (7)$$



8. PUT DOWN-and-OUT

- the price has to decrease to reach the barrier
- canceled when asset price reaches the barrier

$$\Phi_t = \max(X - S_T, 0) \quad \text{if} \quad \min_{0 \leq t \leq T} (S_t) \geq L \quad (8)$$

